Analysis Report: Model-Free vs Model-Based Reinforcement Learning

This report analyzes the behavioral difference between Model-Free and Model-Based Reinforcement Learning using a simple two-stage environment. The task demonstrates how standard Q-Learning can incorrectly assign credit to an action after a rare rewarded transition, while a Model-Based agent correctly reasons about the environment structure.

1. Environment Structure

The environment contains one initial state S0 and two possible actions: A1 and A2. The transition dynamics are probabilistic.

Current State	Action	Next State Probabilities
S0	A1	0.8 → S1, 0.2 → S2
S0	A2	0.2 → S1, 0.8 → S2

Reward probabilities: State S1 gives reward with probability **0.8** — State S2 gives reward with probability **0.2**

Therefore, the optimal long-term strategy is to choose **A1** because it usually leads to the more rewarding state S1.

2. Mathematical Blindspot of Q-Learning

The standard Bellman update equation for Q-Learning is:

```
Q(s,a) <- Q(s,a) + alpha[r + gamma * max Q(s',a') - Q(s,a)]
```

where:

- alpha is the learning rate
- gamma is the discount factor
- r is the received reward

The Model-Free Q-Learning agent only learns from experienced trajectories. It does not explicitly understand transition probabilities or the causal structure of the environment. Suppose a rare event occurs: S0 → A2 → S1 → Reward. Even though A2 usually leads to S2, the received reward is propagated backward to Q(S0,A2). The agent incorrectly increases preference for A2 because the Bellman update only associates actions with obtained rewards.

The problematic term is: r + gamma * max Q(s',a'). This term directly credits the previously selected action for the observed reward without considering whether the transition itself was rare. Thus, the Model-Free agent is mathematically blind to the hidden structure: Action → State → Reward. Instead, it learns only: Action → Reward

3. Model-Based Reinforcement Learning

A Model-Based agent explicitly stores a model of the environment. It must store:

1. Transition probabilities: P(s'|s,a)
2. Reward function: R(s,a,s')
3. State values: V(s)

The Model-Based action-value equation is:

```
Q_MB(s,a) = SUM_s' P(s'|s,a) [ R(s,a,s') + gamma * V(s') ]
```

For the given environment:

Q_MB(S0,A1) = 0.8*V(S1) + 0.2*V(S2)

Q_MB(S0,A2) = 0.2*V(S1) + 0.8*V(S2)

Since V(S1) > V(S2) (reward probability 0.8 vs 0.2), the Model-Based agent correctly concludes that A1 is the better action because it more reliably reaches the high-reward state S1.

4. Hybrid Biological System

Research in neuroscience suggests that biological brains combine both habitual (Model-Free) and planning-based (Model-Based) systems. A simple hybrid formulation is:

```
Q_Hybrid(s,a) = (1 - lambda) * Q_MF(s,a) + lambda * Q_MB(s,a)
```

where:

- Q_MF is the Model-Free value
- Q_MB is the Model-Based value
- lambda controls the balance between the two systems

Interpretation:

- lambda = 0 → purely habitual behavior
- lambda = 1 → purely planning-based behavior
- 0 < lambda < 1 → hybrid biological decision making

5. Conclusion

This experiment demonstrates the conceptual limitation of standard Q-Learning. A Model-Free agent incorrectly assigns credit to actions after rare rewarding outcomes because it lacks an internal model of the environment. In contrast, a Model-Based agent reasons about transition probabilities and future state outcomes, allowing it to make more rational decisions. Biological systems likely combine both approaches, balancing fast habitual learning with flexible planning-based reasoning.